

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 6. Loops - Control Flow

Lecture: 61. Challenge - Fibonacci Series

Section 1: Lecture Summary

This lecture explains the Fibonacci series and how to generate its terms programmatically using a `for` loop in Python. The Fibonacci series starts with two initial terms: 0 (zeroth term) and 1 (first term). Each subsequent term is the sum of the previous two terms. The lecture covers the mathematical definition of Fibonacci terms, demonstrates step-by-step how to compute terms using addition, and then translates this logic into a Python program. The key focus is on understanding the iterative approach with variables representing the series terms, updating these variables inside a loop, and outputting the nth Fibonacci term. The lecture emphasizes practicing this program multiple times to develop programming skills and strong familiarity.

Section 2: Key Concepts and Explanations

- Fibonacci Series Definition

- A sequence where each number is the sum of the two preceding ones.
- Starts with:
 - $\text{fib}(0) = 0$ (zeroth term)
 - $\text{fib}(1) = 1$ (first term)
- For $(n > 1)$,
$$\text{fib}(n) = \text{fib}(n-1) + \text{fib}(n-2)$$

\]

- Terms and Indexing

- Terms are zero-indexed starting at 0. For example:

n	0	1	2	3	4	5	6	...	10
fib(n)	0	1	1	2	3	5	8	...	55

- Iterative Calculation Approach

- Initialize two variables representing the first two terms:
 - `a = 0` (zeroth term)
 - `b = 1` (first term)
- Calculate the next term by adding these two (`c = a + b`)
- Update `a` to `b`, and `b` to `c` to move forward in the series
- Repeat this process `n` times to reach the `n`th term

- For Loop Usage

- Use a `for` loop to perform the addition and updates repeatedly
- Each iteration corresponds to computing the next term in sequence

- Output

- After `n` iterations, the variable `a` contains the `n`th Fibonacci number

- Programming Skills Advice

- Practice the same program multiple times to reinforce understanding and fluency
- Learning programming demands both concept comprehension and repeated skill practice
- Mastery of small programs builds confidence for more complex challenges later

Section 3: Example Code and Use Cases

Program to find the nth Fibonacci number using a for loop

```
n = int(input("Enter the term number you want in Fibonacci series: "))

a = 0 # Zeroth term
b = 1 # First term

for i in range(n):
    c = a + b # Next term
    a = b     # Update a to previous b
    b = c     # Update b to new term c

print(a) # a now holds the nth Fibonacci term
```

Explanation:

- The program takes an integer input `n` for the desired term number.
- Starts with `a` (fib(0)) = 0 and `b` (fib(1)) = 1.
- For `n` iterations, it computes the next term `c` as the sum of `a` and `b`.
- Updates `a` and `b` to move forward in the series progressively.
- After the loop, `a` contains the nth Fibonacci number and prints it.

Use Cases:

- Input `6` will output `8` (the sixth Fibonacci number).
- Input `8` will output `21` (the eighth Fibonacci number).

Section 4: Key Takeaways

- The Fibonacci series starts with 0 and 1 as the first two terms.
- Every new term is the sum of the two preceding terms.
- The series is zero-indexed: $(\text{fib}(0) = 0, \text{fib}(1) = 1)$.
- An iterative method using two variables and a `for` loop efficiently computes the nth term.
- Variable `a` tracks the current Fibonacci term, while `b` tracks the next term to come.
- Code repetition and practice are crucial to build programming fluency.
- Being able to write this Fibonacci code quickly signifies a solid grasp of loops and variable updates in Python.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Challenge : for Loop

Challenge : for Loop

- Fibonacci Series

Fibonacci Series

Challenge

Fibonacci Series

Challenge

n

012345678910

fib(n)

011235813213455

Fibonacci Series

Challenge

n

0

1

fib(n)

0

1

Fibonacci Series

Challenge

n

0

1

2

fib(n)

0

1

1

Fibonacci Series

Challenge

n

0	1	2	3
---	---	---	---

fib(n)

0	1	1	2
---	---	---	---

Fibonacci Series

Challenge

n

0	1	2	3	4
---	---	---	---	---

fib(n)

0	1	1	2	3
---	---	---	---	---

Fibonacci Series

Challenge

n

0	1	2	3	4	5
---	---	---	---	---	---

fib(n)

0	1	1	2	3	5
---	---	---	---	---	---

Fibonacci Series

Challenge

n

0	1	2	3	4	5	6
---	---	---	---	---	---	---

fib(n)

0	1	1	2	3	5	8
---	---	---	---	---	---	---

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7
fib(n)	0	1	1	2	3	5	8	13

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8
fib(n)	0	1	1	2	3	5	8	13	21

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9
fib(n)	0	1	1	2	3	5	8	13	21	34

Fibonacci Series

Challenge

n

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

fib(n)

0	1	1	2	3	5	8	13	21	34	55
---	---	---	---	---	---	---	----	----	----	----

fib(0) = 0

Fibonacci Series

Challenge

n

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

fib(n)

0	1	1	2	3	5	8	13	21	34	55
---	---	---	---	---	---	---	----	----	----	----

fib(0) = 0

fib(1) = 1

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

fib(0) = 0

fib(1) = 1

fib(n) =

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

fib(0) = 0

fib(1) = 1

fib(n) = fib(n-1)

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

fib(0) = 0

fib(1) = 1

fib(n) = fib(n-1) + fib(n-2)

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

n = 6

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

n = 6

fib(6) = 8

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

n = 8

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

n = 8

fib(8) = 21

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

b

0

1

n = 0

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

b

c

0

1

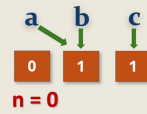
0+1

n = 0

Fibonacci Series

Challenge

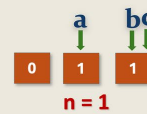
n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55



Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55



Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

b

c

0

1

1

1+1

n = 1

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

b

c

0

1

1

2

n = 1

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

bc

0112

n = 2

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

b

c

01121+2

n = 2

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

b

c

0

1

1

2

3

n = 2

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

b

c

0

1

1

2

3

n = 2

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55



Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

bc

0

1

1

2

3

n = 3

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

bc

0

1

1

2

3

n = 3

a = 0

b = 1

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

bc

01123

n = 3

a = 0

b = 1

for i in range(0, n):

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

bc

01123

n = 3

a = 0

b = 1

for i in range(0, n):

c = a + b

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

bc

01123

n = 3

```
a = 0
b = 1
for i in range(0, n):
    c = a + b
    a = b
```

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

a

bc

01123

n = 3

```
a = 0
b = 1
for i in range(0, n):
    c = a + b
    a = b
    b = c
```

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

0

1

1

2

3

a

bc

n = 3

```
n = int( input ('Enter a Number'))
a = 0
b = 1
for i in range(0 , n ):
    c = a + b
    a = b
    b = c
```

Fibonacci Series

Challenge

n	0	1	2	3	4	5	6	7	8	9	10
fib(n)	0	1	1	2	3	5	8	13	21	34	55

0

1

1

2

3

a

bc

n = 3

```
n = int( input ('Enter a Number'))
a = 0
b = 1
for i in range(0 , n ):
    c = a + b
    a = b
    b = c
print( a )
```

